

Asymmetric Predictive Testing for Aging in SRAMs

Yunkun Lin

Department of Electrical Engineering
University of Southern California
yunkunli@usc.edu

Mingye Li

Department of Electrical Engineering
University of Southern California
mingyel@usc.edu

Sandeep Gupta

Department of Electrical Engineering
University of Southern California
sandeep@usc.edu

Abstract—To avoid corruption of user data, predictive testing methods have been proposed to identify SRAMs likely to fail in the near future due to aging. These methods use aggressive operating conditions, e.g., adjustments to wordline voltages or the power supply voltage, that are calibrated to provide high coverage of SRAMs likely to fail in the near future, but end up with some over-testing, i.e., spuriously identifying some chips as likely to fail.

We first present our study which discovered that a large fraction of over-tested chips fail due to read faults triggered during read-1 operations. Our analysis identifies asymmetric aging in SRAM cells, which are more likely to store zeros, as the root cause for this. We build on this discovery to propose an *asymmetric predictive testing method* which performs writes using normal voltages, read-0 at aggressive voltages, and read-1 at less aggressive voltages. We demonstrate that this method significantly reduces over-testing by over $3\times$ to $5\times$, for low limits on under-testing. We also propose and use a new statistical sampling and simulation method to enable fast convergence and accurate evaluation of asymmetric predictive testing.

Index Terms—Predictive tests, asymmetric predictive tests, SRAMs, process variation, over-testing, under-testing, aging, BTI, importance sampling, Monte Carlo simulations

I. INTRODUCTION

Intel has reported an aging-induced $V_{\min}$ shift instability issue in its 13th and 14th Gen Core desktop processors [1]. Similarly, Google researchers in [2] highlighted significantly higher-than-expected failure rates in data center processors, identifying aging as a key factor contributing to these failures.

Static Random Access Memories (SRAMs) dominate processor chip areas. Hence, process variations and aging significantly influence SRAM performance, energy efficiency, and reliability. Process variations have become increasingly significant as feature sizes have decreased. Also, SRAM cells have become more susceptible to progressive degradation over their lifetimes due to aging mechanisms, primarily Bias Temperature Instability (BTI). (Hot Carrier Injection (HCI) is important for logic but less important for SRAMs.) These effects shift threshold voltages, slow down transistors, reduce noise margins, and can lead to failures. The cumulative effects of process variations and aging can result in a dramatic reduction in SRAM performance and reliability.

Prior research: Improving performance and resilience in the presence of process variations and aging effects has been an active study area [3]–[19]. Approaches that integrate sensors for real-time monitoring of SRAMs and adjusting their operating modes are outlined in [15]–[19]. Design modifications aimed at improving aging resilience are discussed in [9], [20], while techniques for managing operational modes,

including periodic recovery phases, appear in [21], [22]. Post-manufacturing predictive tests to filter out chips prone to aging-induced failure are proposed in [3]–[8].

In this paper, we analyze the data regarding existing predictive tests, especially *under-testing*, i.e., the failure to reject chips that will fail due to future aging thereby potentially corrupting user data, and *over-testing*, i.e., the unnecessary rejection of chips that will remain functional despite future aging. We discover that a very large proportion of over-testing occurs due to errors provoked by read-1 operations. We further identify that this is rooted in the distribution of data values stored — zeros stored much more often than ones [23] — which causes asymmetric aging.

Based on this discovery, we propose a new asymmetric predictive testing method to further optimize the balance between over-testing and under-testing. This new method addresses the combined effects of process variations and aging and significantly reduces over-testing compared to prior predictive testing approaches, while ensuring that under-testing remains below a user-specified limit. Importantly, this new method achieves this without requiring complex SRAM redesign or active management of operational modes required by above prior research.

The evaluation of the effectiveness of our new predictive tests necessitates a new statistical sampling approach. We develop enhanced methods for importance sampling (IS) Monte Carlo (MC) simulations to accurately estimate yield, over-testing, and under-testing. This advances previous IS-MC methods by incorporating models of SRAM failures. We use this new IS-MC method to demonstrate that, for the same under-testing constraints, the asymmetric approach significantly reduces over-testing, compared to previous predictive testing methods [3]–[8]. We also demonstrate that the new IS-MC method achieves faster convergence and greater accuracy.

In Section II, we analyze the primary causes of over-testing and introduce our asymmetric predictive testing method. Section III summarizes the IS-MC simulation approach and demonstrates the effectiveness of our proposed predictive testing and simulation techniques. Finally, we conclude with a summary and outline directions for future research.

II. PREDICTIVE TESTS TO MITIGATE AGING IN SRAMs

To address the aging problem in SRAMs, which worsens over time, our goal is to develop predictive tests for aging-induced faults to identify SRAMs that will fail with the level of aging expected with usage in the near future, i.e., over the next m months. The goal is to use the test results to take preemptive actions that help avoid corruption of user data due to aging failures.

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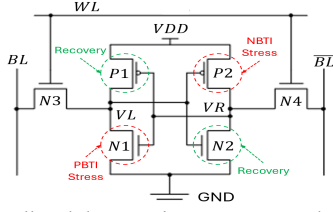


Fig. 1. 6T SRAM cell and the stress/recovery status when “0” is stored [24]

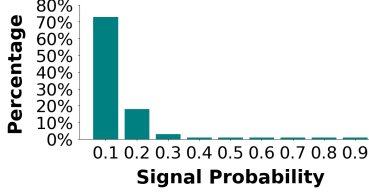


Fig. 2. The signal probability distribution of SRAM cells extracted from [23]

A. Background on SRAMs

(a) In this paper, we use as a case study a 32KByte SRAM array along with the 6T SRAM cell design, un-correlated Gaussian process variation model, a BTI aging model, post manufacturing yield, and aging induced failure rate (under-testing) computation methods from [3], [4], [25]–[27]. Our selection of this 32KByte SRAM array is motivated by its practicality. On one hand this size is widely used in low-level cache implementations, making it realistic as well as sufficiently large to capture all key features that affect predictive testing under the combined effects of process variation and aging. On the other hand, it is small enough to permit extensive simulation studies required to discover the key underlying trends and to thoroughly evaluate new methods.

(b) For estimating BTI aging, we use a real workload where the signal probability distribution in data caches is extracted [23]. This distribution is shown in Figure 2, where the signal probability is the probability that an SRAM cell stores the value “1” over its lifetime.

This distributions shows that SRAM cells that store data in processors are significantly more likely to store “0”. When the 6T cell holds a “0”, as illustrated in Figure 1, the N1 and P2 transistors experience stress while the N2 and P1 transistors undergo recovery. Consequently, a higher likelihood of the cell being in the “0” state results in a greater $|\Delta V_{th}|$ due to aging for the N1 and P2 transistors compared to N2 and P1.

(c) For fault model, we focus on *aging-induced read delay faults* [28] and *read destructive faults* [29]. A read delay fault occurs where there is insufficient time for bit or bit-bar line to discharge before the output of sense amplifier is outputted. A read destructive fault occurs when the value stored in a cell is flipped during a read operation.

B. Background on predictive testing for aging in SRAMs

Several predictive testing methods targeting read delay and destructive faults have been proposed in [3]–[8]. A primary criterion for selecting the most effective approaches is the balance they achieve between under-testing and over-testing to predict the aging failures in the future. Figure 3 illustrates this important guiding principle. Here, under-testing represents SRAMs that pass our predictive tests but fail after aging, and over-testing represents SRAMs that fail our predictive tests but still work after actual future aging.

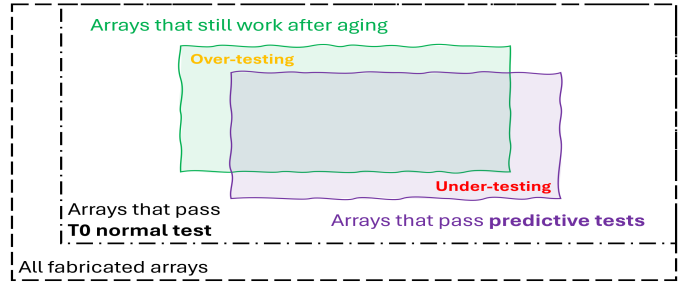


Fig. 3. A Venn diagram explaining the correlation between the post manufacturing (T0) normal test, predictive test, arrays that work after aging

Our ideal goal is to design predictive tests that achieve zero under-testing *and* zero over-testing. However, such a precise control during predictive testing is not possible since we only adjust one or two key voltages and/or frequencies during predictive testing. Hence, our practical objective is the following constrained-optimization.

Objective: Minimize total over-testing.

Constraint: Total under-testing $\leq \alpha$.

The minimization of over-testing maximizes a chip vendor’s revenue while the constraint on under-testing is determined by the part-per-million (ppm) failure rates promised to customers by the vendor. For chip users, the maximum value of total under-testing, α , determines the probability of failures. In turn, this determines the probability of corruption of their data.

In our prior work [3] [4], the trade-offs between over-testing and under-testing are discussed for the **wordline over-drive tests** for predicting read destructive faults and **wordline under-drive tests** for predicting read delay faults.

We start this research by studying the above predictive tests using our improved IS-MC method, presented in Section III. To gain deeper insights into the root causes of over- and under-testing, we partitioned the data into finer categories. The most informative results are summarized in Table I for predicting read destructive faults and Table II for read delay faults.

Both these tables separately show under-testing and over-testing that occur during read-0 and read-1 operations for the prediction of 1-month aging under the skewed workload [23]. These tables do not show the counts for under-testing and over-testing of both read-0 and read-1 failures occurring in the same cell as these values are negligible. These tables show how over-testing varies with different wordline voltage adjustments. Our goal is to identify the root cause of over-testing in predictive testing and explore ways to mitigate.

C. Root cause of large over-testing given under-testing constrain for predictive testing - Read destructive faults

We start by studying the aging induced read destructive faults predicted by wordline over-drive tests, where a higher wordline voltage V_{WL} is applied, such that $V_{WL} = V_{WL,nominal} + \Delta V_{WL}$ during testing. This provides the following two observations.

Observation-1: As shown in Rows 2-4 of Table I, read-0 destructive faults after aging dominate the under-testing of read destructive faults tested by wordline over-drive, accounting for nearly all of the 3.48% aging-induced failure rate when predictive tests are not applied (see column 2 rows 2-4).

TABLE I
WORDLINE OVER-DRIVE [3] [4] PREDICTING READ DESTRUCTIVE FAULTS: OVER-TESTING AND UNDER-TESTING OF READ-0 AND READ-1 FOR 1-MONTH AGING UNDER THE SKEWED WORKLOAD [23], FOR VARIOUS WORDLINE OVERDRIVE VOLTAGES

Wordline over-drive (ΔV_{WL}) value	0mV	2mV	4mV	6mV	8mV	10mV	12mV	14mV	16mV	18mV	20mV	21mV	22mV	23mV	24mV	25mV	26mV
Total Under-testing	3.48%	2.87%	2.57%	2.25%	1.95%	1.70%	1.41%	1.21%	0.90%	0.70%	0.51%	0.43%	0.33%	0.25%	0.19%	0.13%	0.08%
Read-0 only under-testing	3.48%	2.87%	2.57%	2.25%	1.95%	1.70%	1.41%	1.21%	0.90%	0.70%	0.51%	0.43%	0.33%	0.25%	0.19%	0.13%	0.08%
Read-1 only under-testing	≈ 0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Total Over-testing	0	0.06%	0.14%	0.22%	0.31%	0.41%	0.55%	0.67%	0.86%	1.07%	1.29%	1.35%	1.54%	1.62%	1.79%	1.92%	2.12%
Read-0 only over-testing	0	≈ 0	≈ 0	≈ 0	0.01%	0.03%	0.06%	0.08%	0.15%	0.20%	0.31%	0.35%	0.42%	0.48%	0.53%	0.59%	0.67%
Read-1 only over-testing	0	0.06%	0.14%	0.22%	0.30%	0.38%	0.49%	0.59%	0.71%	0.87%	0.98%	1.05%	1.12%	1.17%	1.26%	1.33%	1.45%

TABLE II
WORDLINE UNDER-DRIVE [3] PREDICTING READ DELAY FAULTS: OVER-TESTING AND UNDER-TESTING OF READ-0 AND READ-1 FOR 1-MONTH AGING UNDER THE SKEWED WORKLOAD [23], FOR VARIOUS WORDLINE UNDERDRIVE VOLTAGES

Wordline under-drive ($-\Delta V_{WL}$) value	0mV	1mV	1.5mV	2mV	2.5mV	3mV	3.5mV
Total Under-testing	1.87%	1.11%	0.63%	0.30%	0.15%	0.06%	0.02%
Read-0 only under-testing	1.17%	0.79%	0.54%	0.27%	0.15%	0.06%	0.02%
Read-1 only under-testing	0.70%	0.32%	0.09%	0.03%	≈ 0	0	0
Total Over-testing	0	≈ 0	0.05%	0.25%	0.60%	1.06%	1.55%
Read-0 only over-testing	0	0	≈ 0	0.03%	0.14%	0.33%	0.55%
Read-1 only over-testing	0	0	0.05%	0.22%	0.46%	0.72%	1.00%

Observation-2: As shown in rows 5–8 of Table I, read-1 related over-testing dominates the overall over-testing, for example, with an under-testing threshold of less than 0.2%, we need to select $\Delta WL = 24mV$ wordline over-drive, and the over-testing rate for read-0 faults is 0.53%, compared to 1.26% for read-1 faults.

The above observations from failure data for existing predictive testing caused us to investigate their root cause(s). This provided the hypothesis that the asymmetry in stored data leads to asymmetric aging effects, which in turn contribute to the observed distribution of over-testing and under-testing for aging predictive testing.

Analysis to validate this hypothesis: Table I models SRAM aging based on real-world SRAM usage, incorporating the skewed workload extracted from traces for actual programs [23]. As shown in Fig. 2, this distribution is dominated by stored '0', which causes PMOS P2 to suffer more from NBTI aging compared to PMOS P1, and NMOS N1 to suffer more from PBTI aging compared to NMOS N2 as shown in Figure 1 and explained in Section II-A.

Consider read destructive faults caused by such aging, as shown in Fig 1, where 6T SRAM cell stores 0 during a read operation. After the bitlines are precharged and the wordline is activated, the voltage division between N3 and N1 determines the voltage of V_L during the read operation. Due to aging, N1 is weakened by PBTI, resulting in a higher voltage at V_L . At the same time, the trip voltage V_{trip} (the minimum input voltage at which an inverter's output falls) of the inverter consisting of N2 and P2 decreases as the strength of P2 is weakened by NBTI. Such higher V_L and lower V_{trip} together make it easier to flip the cell during the read operation, which corrupts the stored value. Hence, the asymmetric aging increases the probability of exciting the read destructive faults during read-0 operations.

In contrast, a similar analysis when the cell stores a '1' shows that the above shifts in threshold voltages due to asymmetric aging also *increases the probability of read destructive faults during read-1 operations, but by a much smaller amount*

relative to the increase for read-0.

As can be seen in the above tables, stricter stress conditions during predictive testing, namely higher ΔV_{WL} , increases the ability of the predictive tests to detect faults due to future aging. However, for the same duration of future aging, asymmetric aging increases the severity of aging faults excited by read-0 much more than faults excited by read-1. Hence, the level of over-drive voltage required to satisfy the given under-testing constraint for read-0 excited faults causes significant over-testing of read-1 excited faults.

Hence, the above shows that due to asymmetric aging, read-0 destructive faults dominate under-testing, consistent with *Observation-1*, and read-1 destructive faults dominate over-testing, consistent with *Observation-2*. Simply, by analyzing wordline over-drive over-testing results and conducting circuit analysis, we have identified the root cause of over-testing in predictive testing for aging-induced read destructive faults.

D. Root cause of large over-testing given under-testing constrain for predictive testing - Read delay faults

We carried out a similar study for aging-induced read delay faults as predicted by the wordline under-drive test, where a reduced wordline voltage V_{WL} is used. As shown in Table II, we observe similar trends as for read destructive faults: aged read-0 delay faults dominate under-testing, while read-1 delay faults dominate over-testing tested by wordline under-drive.

We undertake a similar circuit analysis as above. After aging, PBTI aging degrades the pull-down transistor N1 which slows the discharge rate for read-0, causing read-0 related delay faults. Similarly, aging in N2 slows the discharge rate for read-1. Due to the skewed workload, where '0' is stored more frequently, N1 experiences greater aging compared to N2. Consequently, read-0 delay faults are more severe than read-1 delay faults.

Read delay faults are different from read destructive faults in one way, the read delay faults are less severe compared to read destructive faults. This is evident in the magnitudes of under-testing when $\Delta V_{WL} = 0$ in Tables II and I, namely 1.87% and 3.48% respectively for read delay and read destructive faults. This difference arises because read delay faults are solely caused by PBTI aging in the two pull-down NMOS transistors, N1 and N2, whereas read destructive faults are caused by both PBTI aging in N1 and N2 and NBTI aging in the two pull-up PMOS transistors, P1 and P2.

E. Proposed asymmetric predictive testing method - Read destructive faults

Our goal is to refine predictive testing methods to reduce over-testing while ensuring that we still satisfy the constraints on under-testing.

TABLE III

MARCH TEST FOR **SYMMETRIC** WORDLINE OVER-DRIVE PREDICTING AGING-INDUCED READ DESTRUCTIVE FAULTS

Operation	$\uparrow w0$	$\uparrow r0$	$\uparrow w1$	$\uparrow r1$
ΔV_{WL}	0	$\Delta V_{WL,A}$	0	$\Delta V_{WL,A}$
Positive/Negative	0	Positive	0	Positive

TABLE IV

MARCH TEST FOR **ASYMMETRIC** WORDLINE OVER-DRIVE PREDICTING AGING-INDUCED READ DESTRUCTIVE FAULTS

Operation	$\uparrow w0$	$\uparrow r0$	$\uparrow w1$	$\uparrow r1$
ΔV_{WL}	0	$\Delta V_{WL,C}$	0	$\Delta V_{WL,D}$
Positive/Negative	0	Positive	0	Positive

Asymmetric predictive test: The existing predictive tests increase wordline over-drive voltage, ΔV_{WL} , until the under-testing, primarily of read-0 excited faults, falls below the required limit, α . However, by the time we reach the appropriate value of ΔV_{WL} , these tests cause high over-testing of read-1 excited faults. Table III illustrates such a test where the over-drive voltage $\Delta V_{WL,A}$ is used for read-0 as well as read-1.

Key idea-1: Use different over-drive voltages during read-0 and read-1 operations. In particular, use a value of ΔV_{WL} , for read-0 operations to ensure low under-testing of read-0 excited faults, while using a lower value of ΔV_{WL} for read-1 operations to reduce over-testing of read-1 excited faults.

Such an *asymmetric predictive test* is illustrated in Table IV using over-drive voltage $\Delta V_{WL,C}$ during read-0 operations and $\Delta V_{WL,D}$ during read-1 operations, where $\Delta V_{WL,C} > \Delta V_{WL,D}$. This approach allows us to accurately predict read-0 faults by using a sufficiently high over-drive voltage $\Delta V_{WL,C}$ during read-0, while avoiding the additional over-testing for read-1 faults by using lower over-drive voltage over-drive voltage $\Delta V_{WL,D}$ during read-1.

Avoiding over-testing at half-selected cells: The goal of asymmetric predictive testing is to avoid over-testing, especially of read-1 excited read destructive faults. The above approach reduces such over-testing by using a lower over-drive voltages, $\Delta V_{WL,D}$ during read-1 operations (r1 in Table IV).

However, most SRAM array include multiple words per row. For example, the SRAM we are using as a case-study contains 8 64-bit words per row. In any SRAM with two or more words per row, during each write or read operation on one word inherently triggers a read operation (without data output) on the remaining words in that row. The cells in these other words are referred to as half-selected cells, as illustrated in Figure 4. Specifically, when a write operation is performed on a word, every other cell in that row is half-selected and a read operation is performed at each such cell. Further, each such half-selected cell executes a read-0 or a read-1 operation, depending on whether it was storing a 0 or a 1 prior to the operation.

Key idea-2: To avoid over-testing of read-1 excited faults, in our asymmetric tests we use normal wordline voltage, i.e., $\Delta V_{WL} = 0$, during all write operations, as shown in Table IV.

The use of zero over-/under-drive during all write operations, avoids any over-testing of read-0 and read-1 excited faults in cells that are half-selected during any write operation.

Key idea-3: The final issue that we address is about the read operations applied at half-selected cells during read operations. Next, we show how our wordline over-drive tests in Table IV

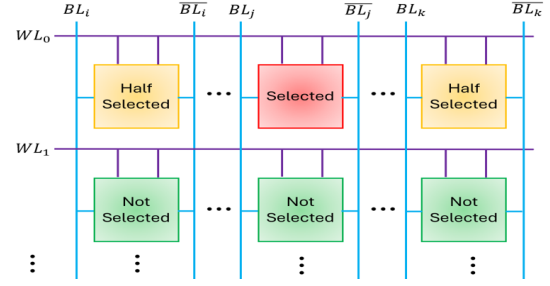


Fig. 4. Selected and half-selected SRAM cells during read/write operations avoid any over-testing in half-selected cells **and** guarantee the detection of all aging-induced read destructive faults.

First, given the structure of the march test in Table IV, when a read-0 is performed on a cell, all the half-selected cells also contain '0' and hence the read operation performed on the half-selected cells is also read-0. Similarly, when a read-1 is performed on a cell, the operation performed at every half-selected cell is also read-1. Hence, all such half-selected reads are performed using the appropriate level of over-drive voltage and no additional over-testing occurs.

Second, when a stressed read operation is performed on a specific word, the wordline over-drive voltage may trigger an aging-induced read failure in a half-selected cell. In such a case, the erroneous value caused by the read destructive fault is stored in the corresponding cell, say C_{hs} . Further, since a single cell cannot exhibit both a read-0 and a read-1 destructive faults simultaneously (these fault types are mutually exclusive within distinct regions of V_{th} , as explained in Section III), our march test can safely continue with read operations on the remaining words in the same row without overwriting the above mentioned erroneous value capture in C_{hs} . Eventually, when we perform a read operation on cell C_{hs} , we read out the erroneous value stored when the read destructive fault was excited earlier, thereby detecting that fault.

In summary, our asymmetric predictive test method meets all our goals, and is summarized in Table IV. The first row of the table shows the operations to be applied sequentially to all words within each row and across all rows in the array. The last two rows specify the stress conditions during these operations for predictive testing. In our new asymmetric predictive testing, we use zero over-drive during all write operations and use distinct over-drive conditions for predictive testing during read-0 and read-1 operations, i.e., use $\Delta V_{WL,C}$ for read-0 and $\Delta V_{WL,D}$ for read-1 to enable asymmetric wordline over-drive testing, where $\Delta V_{WL,C} > \Delta V_{WL,D}$.

To determine the optimal stress conditions ($\Delta V_{WL,C}$ and $\Delta V_{WL,D}$) for predicting 1-month aged read destructive faults, we need to solve a constrained optimization problem. This requires ensuring that the total increase in under-testing for read-0 as well as read-1 related destructive faults remains below the threshold α , while minimizing the over-testing for both fault types. For under-testing below 0.2%, the optimal values are $\Delta V_{WL,C} = 24$ mV and $\Delta V_{WL,D} \approx 0$ mV, based on the over-testing and under-testing counts shown in Table I, in which the total under-testing is 0.19% while the over-testing is 0.53%. This over-testing is about $3.4\times$ smaller than the symmetric predictive testing which is 1.92%, which uses the

TABLE V
MARCH TEST FOR SYMMETRIC WORDLINE UNDER-DRIVE PREDICTING
AGING-INDUCED READ DELAY FAULTS

Operation	$\Downarrow w0$	$\Downarrow r0$	$\Downarrow w1$	$\Downarrow r1$
ΔV_{WL}	0	$\Delta V_{WL,B}$	0	$\Delta V_{WL,B}$
Positive/Negative	0	Negative	0	Negative

TABLE VI
MARCH TEST FOR ASYMMETRIC WORDLINE UNDER-DRIVE PREDICTING
AGING-INDUCED READ DELAY FAULTS

Operation	$\Downarrow w0$	$\Downarrow r0$	$\Downarrow w1$	$\Downarrow r1$
ΔV_{WL}	0	$\Delta V_{WL,E}$	0	$\Delta V_{WL,F}$
Positive/Negative	0	Negative	0	Negative

over-drive voltage $\Delta V_{WL,A} = 24$ mV for read-0 and read-1 operations, as shown in Table III.

F. Method to improve the over-testing issues for predictive testing - Read delay faults

As shown in Table V, existing predictive tests, such as wordline under-drive, use the same under-drive voltage $\Delta V_{WL,B}$ for both read-0 and read-1 operations.

Inspired by asymmetric wordline over-drive for read destructive faults, we apply different wordline under-drive voltages for read-0 and read-1 operations to predict read delay faults. Table VI outlines our approach: we use $\Delta V_{WL,E}$ for read-0 to reduce under-testing and $\Delta V_{WL,F}$ for read-1 to reduce over-testing. Both $\Delta V_{WL,E}$ and $\Delta V_{WL,F}$ are negative, with $|\Delta V_{WL,E}| > |\Delta V_{WL,F}|$. During all write operations, zero over-/under-drive is applied to prevent over-testing of read-0 and read-1 faults in half-selected cells.

Key Idea-4: We address the issue of read operations in half-selected cells triggering read delay faults. Table VI shows how our method prevents over-testing in these cells while ensuring the detection of all aging-induced read delay faults.

First, in the march test structure (Table VI), half-selected cells also perform read operations during a cell read. However, faults triggered in half-selected cells do not propagate to outputs, avoiding any over-testing. Second, if a read delay fault is triggered in a half-selected cell, C_{hs} , its stored value remains unchanged, ensuring no over-testing occurs when we perform a real read operation on that cell C_{hs} .

In summary, our asymmetric wordline under-drive method achieves all design goals and is detailed in Table VI, using a similar march test structure as the wordline over-drive method discussed in Section II-E.

Finally, to determine optimal stress conditions $\Delta V_{WL,E}$ for read-0 and $\Delta V_{WL,F}$ for read-1, we need to solve a constrained optimization problem minimizing over-testing while keeping under-testing within a set limit. For under-testing $< 0.2\%$, the optimal values are $\Delta V_{WL,E} = -2.7$ mV and $\Delta V_{WL,F} = -1.5$ mV, resulting in total under-testing of 0.19% and over-testing of 0.2%, a $2.5\times$ reduction compared to symmetric predictive testing ($\sim 0.5\%$).

G. Method to improve the over-testing issues for predictive testing - Read destructive and delay faults

Building on the two approaches discussed above, our goal is to develop a predictive test method targeting read destructive and delay faults after aging, which reduces over-testing while satisfying the under-testing constraints.

TABLE VII
MARCH TESTS FOR ASYMMETRIC WORDLINE OVER- AND UNDER-DRIVE
PREDICTING AGING-INDUCED READ DESTRUCTIVE AND DELAY FAULTS

Operation	$\Downarrow w0$	$\Downarrow r0$	$\Downarrow r0$	$\Downarrow w1$	$\Downarrow r1$	$\Downarrow r1$
ΔV_{WL}	0	$\Delta V_{WL,C}$	$\Delta V_{WL,E}$	0	$\Delta V_{WL,D}$	$\Delta V_{WL,F}$
Positive/Negative	0	Positive	Negative	0	Positive	Negative

Key idea-5: To reduce the predictive test's complexity, we combine the two march tests for asymmetric wordline under-drive and over-drive to form a new march test, in which the write operations are shared by these two test methods.

Because applying two different wordline voltages in the same read operation is not feasible, we apply wordline over-drive and under-drive during two separate read operations. As shown in Table VII, we begin by initializing the SRAMs with a write-0 operation under normal working conditions followed by a wordline over-drive read-0 with positive $\Delta V_{WL,C}$ to predict read-0 destructive faults, and then a wordline under-drive read-0 with negative $\Delta V_{WL,E}$ to predict read-0 delay faults. A similar procedure is followed for read-1 related faults. As we discussed in above, this method avoids extra over-testing and reduces test complexity from $8N$ to $6N$ compared to testing the two types of read faults separately.

To determine the optimal stressing conditions ($\Delta V_{WL,C}$, $\Delta V_{WL,D}$, $\Delta V_{WL,E}$, $\Delta V_{WL,F}$), we need to solve a constrained optimization problem aimed at minimizing total over-testing while keeping under-testing cases below a preset threshold (e.g., 0.2%) for predicting both read faults after aging. As discussed in Section III, the exclusivity of read delay and destructive faults, combined with the exclusive/independence of read-0 and read-1 destructive/delay faults, allows efficient computation of total counts. By summing the over-testing and under-testing values for read-0 and read-1 destructive and delay faults under the given stress conditions, the constrained optimization process is greatly simplified.

III. ENHANCED IMPORTANCE SAMPLING MONTE CARLO (IS-MC) SIMULATION METHOD AND RESULTS

Considering the fact that the failure rate, over-testing, and under-testing counts for a single cell are extremely small (10^{-8}), to address this, we apply IS-MC simulation methods to focus sampling on regions with higher failure likelihood, using weights to correct distribution bias. Due to page limit, we briefly outline our new IS-MC simulation approach.

A. Enhanced IS-MC simulation method in SRAM

Table VIII summarizes the dangerous regions for the six transistors corresponding to each failure mechanism, with '+' and '-' indicating stronger and weaker transistors, respectively, increasing read fault probabilities, and 'N/A' representing no impact. The table reveals that read-0 and read-1 destructive faults have distinct, mutually exclusive dangerous regions, with NMOS transistors N1 and N2 having opposing dangerous region. Similarly, read delay faults are independent, each with a unique dangerous region.

These exclusiveness and independencies create a concave failure rate relationship across the parameter space of the six transistors, rendering convex optimization methods from [30] [31] ineffective for SRAM simulations. To address this, we fit two independent sampling functions $g_{r0}(x)$ and $g_{r1}(x)$,

TABLE VIII

DANGEROUS REGIONS FOR ALL SIX TRANSISTORS IN A 6T SRAM CELL FOR READ DESTRUCTIVE AND DELAY FAULTS, '+' REPRESENTS STRONGER, '-' REPRESENTS WEAKER, 'N/A' REPRESENTS NO INFLUENCE

Fault type	Operation	N1	N2	N3	N4	P1	P2
Read Destructive	Read-0	-	+	+	N/A	N/A	-
	Read-1	+	-	N/A	+	-	N/A
Read Delay	Read-0	-	N/A	-	N/A	N/A	N/A
	Read-1	N/A	-	N/A	-	N/A	N/A

modeled as multidimensional Gaussian distributions for read-0 and read-1 faults' dangerous regions, respectively. Finally, The combined sampling function for IS-MC simulations is defined as: $\frac{1}{2}(g_{r0}(x) + g_{r1}(x))$.

B. Experimental results for our asymmetric predictive testing

Using this new IS-MC simulation method, we analyze the impact of over-testing and under-testing by comparing results from our new asymmetric stress approach with the traditional symmetric stress method for read-0 and read-1.

Figures 5 and 6 illustrate the over-testing versus under-testing performance of two predictive testing methods for read destructive and delay faults over a 1-month aging period. The x-axis represents under-testing, and the y-axis represents over-testing. The red curve corresponds to the traditional symmetric stress method (Table III), while the green curve represents our proposed asymmetric stress approach.

As shown in the Figure 5, the new asymmetric stress method for read-0 and read-1 destructive faults performs better than the traditional symmetric approach by significantly reducing over-testing for identical under-testing constraints. For example, when we limit under-testing to below 0.2%, the asymmetric method achieves only 0.53% over-testing for a 1-month aging prediction, compared to 1.79% with the symmetric method. Similarly, as Figure 6 shows for read delay faults, the asymmetric method results in just 0.19% over-testing, whereas

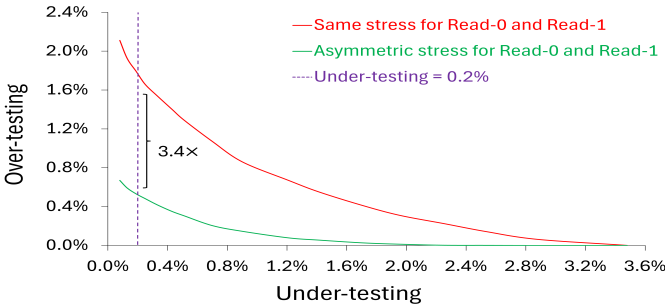


Fig. 5. Over-testing vs Under-testing for 1-month different predictive testing methods for read destructive faults based on methods in Table III and IV

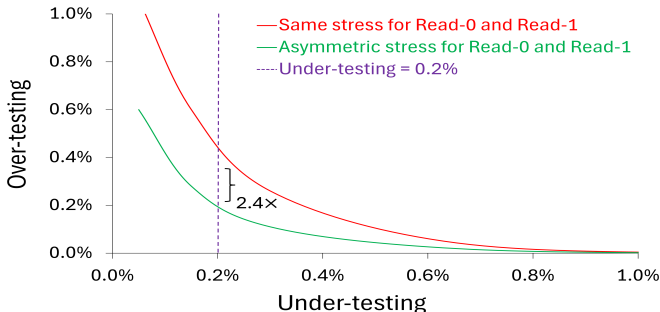


Fig. 6. Over-testing vs Under-testing for 1-month different predictive testing methods for read delay faults based on methods in Table V and VI

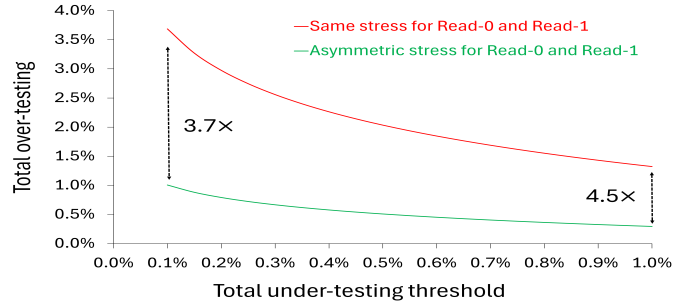


Fig. 7. Minimum over-testing for under-testing threshold (0.1% – 1.0%) for 1-month different predictive testing methods targeting both read destructive and delay faults based on method in Table VII

the symmetric approach has 0.46% over-testing. *These results demonstrate the superior efficiency of the asymmetric method for separately targeting read destructive and delay faults.*

Figure 7 compares asymmetric and symmetric predictive testing methods for read destructive and delay faults induced by 1-month aging. The x-axis represents the under-testing threshold (0.1% to 1%), while the y-axis shows the total over-testing count. As illustrated, the asymmetric predictive testing method achieves a 3x-5x reduction in over-testing at low under-testing thresholds, demonstrating its efficiency.

C. IS-MC simulation results

We analyzed the confidence ratio R_c for over-testing as a function of sample size, comparing our proposed IS-MC simulation method with the previous approach from [30] [31], where lower R_c values correspond to higher accuracy.

The **previous IS-MC method** consistently exhibits higher R_c values, indicating slower convergence and reduced precision. In contrast, the **enhanced IS-MC method** achieves significantly lower R_c values, reflecting faster convergence and greater accuracy. For example, at a sample size of 100,000, the enhanced method reaches an R_c of approximately 0.03, compared to about 0.15 for the previous method — 5x higher. Notably, while the previous method struggles to meet the typical IS-MC convergence threshold of $R_c = 0.05$, our enhanced method comfortably satisfies this criterion.

These results highlight the significant advantages of the proposed IS-MC method in achieving faster convergence and greater accuracy, demonstrating its potential as an efficient improvement over traditional methods.

IV. CONCLUSION AND FUTURE RESEARCH

In this research, we present an enhanced predictive testing method designed to effectively mitigate over-testing when predicting failures caused by asymmetric aging effects. Using a proposed IS-MC simulation method, we demonstrate the efficiency and accuracy of our enhanced predictive testing approach in capturing aging-induced failure SRAMs.

Our ongoing work focuses on deriving the relationship between aging+ (process variation plus aging effect) and the fundamental trade-offs between over-testing and under-testing. Additionally, we are extending our methods by integrating other predictive testing techniques to further enhance testing efficiency for SRAMs and logic circuits.

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